DBMS LAB 7

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Experiment 7: Perform Nested and Complex Queries

Aim: Mastering Subqueries (IN, NOT IN, EXISTS, NOT EXISTS, ANY, ALL), Correlated Subqueries, Derived Tables (FROM clause), HAVING clauses, and Set Operators (UNION, INTERSECT).

1 [Subquery - IN]: Write an SQL query using an IN subquery to find the full names and countries of players who scored more than 50 runs (NRuns > 50) in any single match.

```
SELECT FName, LName, Country
FROM player
WHERE PlayerID IN (
   SELECT PID
   FROM batting
   WHERE NRuns > 50 );
```

```
1  USE anmol_11;
2
3  SELECT FName, LName, Country
4  FROM player
5  WHERE PlayerID IN (
6      SELECT PID
7      FROM batting
8      WHERE NRuns > 50
9  );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FName	LName	Country
MS	Dhoni	India
Sanath	Jayasuriya	Sri Lanka
Sachin	Tendulkar	India

2 [Subquery - NOT IN]: Retrieve all players (PlayerID, FName, LName, Country) who have NEVER bowled in any match using a NOT IN subquery referencing the Bowling table.

```
SELECT PlayerID, FName, LName, Country
FROM player
WHERE PlayerID NOT IN (
   SELECT PID
   FROM bowling );
```

```sql
USE anmol_11;

SELECT PlayerID, FName, LName, Country
FROM player
WHERE PlayerID NOT IN (
    SELECT PID
    FROM bowling );
```

Result Grid | Filter Rows: | Edit:

PlayerID	FName	LName	Country
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
NULL	NULL	NULL	NULL

player 21

3 [Subquery - EXISTS]: Write an SQL query using EXISTS to list all matches (MatchID, Ground, Date) where at least one player hit 2 or more sixes (Six >= 2).

SELECT M.MatchID, M.Ground, M.Date

FROM odimatch M

WHERE EXISTS (

SELECT 1

FROM batting B

WHERE B.MatchID = M.MatchID

AND B.Six >= 2);

```sql
USE anmol_11;

SELECT M.MatchID, M.Ground, M.Date
FROM odimatch M
WHERE EXISTS (
    SELECT 1
    FROM batting B
    WHERE B.MatchID = M.MatchID
      AND B.Six >= 2 );
```

Result Grid | Filter Rows: | Edit:

MatchID	Ground	Date
2755	Colombo	2008-08-24
2689	Brisbane	2008-03-04
NULL	NULL	NULL

4 [Subquery - NOT EXISTS]: Retrieve all players (PlayerID, FName, LName) for whom NO bowling record exists in the Bowling table using a correlated NOT EXISTS clause.

```
SELECT P.PlayerID, P.FName, P.LName
FROM player P
WHERE NOT EXISTS (
   SELECT 1
   FROM bowling B
   WHERE B.PID = P.PlayerID );
```

```
1  USE anmol_11;
2
3  SELECT P.PlayerID, P.FName, P.LName
4  FROM player P
5  WHERE NOT EXISTS (
6      SELECT 1
7      FROM bowling B
8      WHERE B.PID = P.PlayerID );
```

Result Grid | Filter Rows: | Edit:

PlayerID	FName	LName
21001	Virender	Sehwag
25001	MS	Dhoni
27001	Praveen	Kumar
27002	Ishant	Sharma
89001	Sachin	Tendulkar
90001	Brain	Lara
90002	Anil	Kumble
92001	Shane	Warne
92003	Inzamam	UI-Huq
93002	Heath	Streak
93003	Gary	Kirsten
94004	Stephen	Fleming
95001	Ricky	Ponting
95002	Shaun	Pollock
95003	Jacques	Kallis
96001	Rahul	Dravid
96002	Herschelle	Gibbs
96003	Saurav	Ganguly

5 [Scalar Subquery in WHERE]: Find all players (FName, LName, Yborn) who were born before the average birth year (Yborn < AVG(Yborn)) of all registered players.

```
SELECT FName, LName, Yborn
FROM player
WHERE Yborn < (
   SELECT AVG(Yborn)
   FROM player );
```

```
1  USE anmol_11;
2
3  SELECT FName, LName, Yborn
4  FROM player
5  WHERE Yborn < (
6      SELECT AVG(Yborn)
7      FROM player );
```

Result Grid | Filter Rows:

FName	LName	Yborn
Sachin	Tendulkar	1973
Brain	Lara	1969
Anil	Kumble	1970
Sanath	Jayasuriya	1969
Shane	Warne	1969
Muthiah	Murlitharan	1972
Inzamam	UI-Huq	1970
Heath	Streak	1974
Gary	Kirsten	1967
Chaminda	Vaas	1974
Stephen	Fleming	1973
Ricky	Ponting	1974
Shaun	Pollock	1973
Rahul	Dravid	1973
Herschelle	Gibbs	1974
Saurav	Ganguly	1972
Adam	Gilchrist	1971
Michale	Vaughan	1974

6 [Subquery - ALL Operator]: Write a query to find player batting performances where NRuns is greater than ALL runs scored by players representing 'Australia'.

```
SELECT B.MatchID, B.PID, B.NRuns
FROM batting B
WHERE B.NRuns > ALL (
   SELECT B2.NRuns
   FROM batting B2
   INNER JOIN player P
      ON B2.PID = P.PlayerID
   WHERE P.Country = 'Australia' );
```

```
1  USE anmol_11;
2
3  SELECT B.MatchID, B.PID, B.NRuns
4  FROM batting B
5  WHERE B.NRuns > ALL (
6      SELECT B2.NRuns
7      FROM batting B2
8      INNER JOIN player P
9          ON B2.PID = P.PlayerID
10     WHERE P.Country = 'Australia' );
```

Result Grid | Filter Rows: | Export:

MatchID	PID	NRuns
2755	25001	71
2755	91001	60
2689	89001	91

7 [Subquery - ANY / SOME]: Find all bowling performances where Nwickets is greater than ANY wicket count achieved by bowlers from 'Sri Lanka'.

```
SELECT B.MatchID, B.PID, B.Nwickets
FROM bowling B
WHERE B.Nwickets > ANY (
   SELECT B2.Nwickets
   FROM bowling B2
   INNER JOIN player P
      ON B2.PID = P.PlayerID
   WHERE P.Country = 'Sri Lanka' );
```

```
1  USE anmol_11;
2
3  SELECT B.MatchID, B.PID, B.Nwickets
4  FROM bowling B
5  WHERE B.Nwickets > ANY (
6      SELECT B2.Nwickets
7      FROM bowling B2
8      INNER JOIN player P
9          ON B2.PID = P.PlayerID
10     WHERE P.Country = 'Sri Lanka' );
```

Result Grid | Filter Rows: | Export:

MatchID	PID	Nwickets
2689	99001	1
2689	24001	1
2755	94002	1
2755	92002	1
2755	23001	2
2755	98002	3
2689	98002	1

8 [Correlated Subquery]: Write a correlated subquery to list each player who scored the maximum runs (NRuns) among all players from their own Country.

```
SELECT P.FName, P.LName, P.Country, B.NRuns
FROM player P
JOIN batting B ON P.PlayerID = B.PID
WHERE B.NRuns = (
   SELECT MAX(B2.NRuns)
   FROM batting B2
   JOIN player P2 ON B2.PID = P2.PlayerID
   WHERE P2.Country = P.Country );
```

```sql
USE anmol_11;

SELECT P.FName, P.LName, P.Country, B.NRuns
FROM player P
JOIN batting B ON P.PlayerID = B.PID
WHERE B.NRuns = (
    SELECT MAX(B2.NRuns)
    FROM batting B2
    JOIN player P2 ON B2.PID = P2.PlayerID
    WHERE P2.Country = P.Country );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FName	LName	Country	NRuns
Sanath	Jayasuriya	Sri Lanka	60
Sachin	Tendulkar	India	91
Andrew	Symonds	Australia	42

9 [Nested Subquery with HAVING]: Display players whose average runs per match is strictly greater than the overall average runs scored across all matches in the Batting table.

```sql
SELECT P.PlayerID, P.FName, P.LName,
    AVG(B.NRuns) AS Average_Runs
FROM player P
JOIN batting B ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName
HAVING AVG(B.NRuns) > (
  SELECT AVG(NRuns)
  FROM batting );
```

```sql
USE anmol_11;

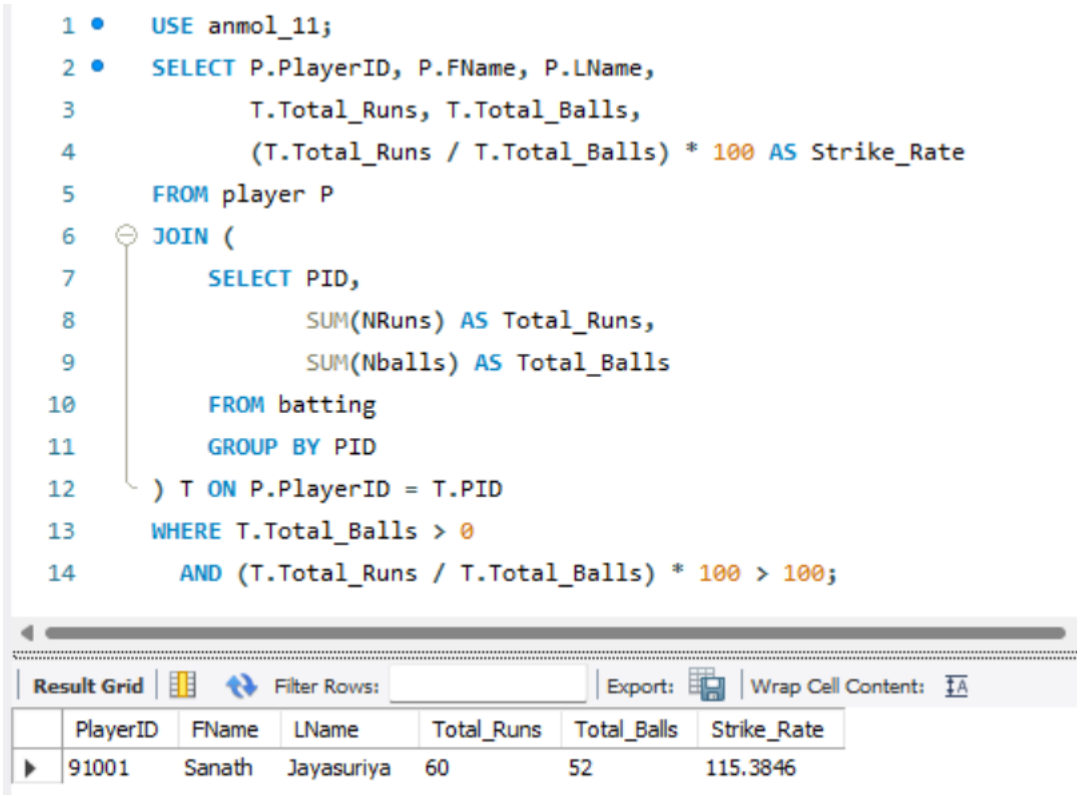
SELECT P.PlayerID, P.FName, P.LName,
       AVG(B.NRuns) AS Average_Runs
FROM player P
JOIN batting B ON P.PlayerID = B.PID
GROUP BY P.PlayerID, P.FName, P.LName
HAVING AVG(B.NRuns) > (
    SELECT AVG(NRuns)
    FROM batting );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerID	FName	LName	Average_Runs
25001	MS	Dhoni	53.5000
91001	Sanath	Jayasuriya	60.0000
89001	Sachin	Tendulkar	91.0000
24001	Andrew	Symonds	42.0000

10 [Derived Table in FROM]: Create a derived table in the FROM clause calculating total runs and total balls faced per player, then select players with a strike rate ((Total_Runs / Total_Balls) * 100) greater than 100.

```
SELECT P.PlayerID, P.FName, P.LName,
       T.Total_Runs, T.Total_Balls,
       (T.Total_Runs / T.Total_Balls) * 100 AS Strike_Rate
FROM player P
JOIN (
   SELECT PID,
          SUM(NRuns) AS Total_Runs,
          SUM(Nballs) AS Total_Balls
   FROM batting
   GROUP BY PID
) T ON P.PlayerID = T.PID
WHERE T.Total_Balls > 0
  AND (T.Total_Runs / T.Total_Balls) * 100 > 100;
```



11 [Set Operator - UNION]: Write a query using UNION to combine the PlayerIDs of players who scored > 50 runs in Batting with players who took >= 3 wickets in Bowling.

```
SELECT PID AS PlayerID
FROM batting
WHERE NRuns > 50
UNION
SELECT PID AS PlayerID
FROM bowling
WHERE Nwickets >= 3;
```

```
1 •  USE anmol_11;
2 •  SELECT PID AS PlayerID
3    FROM batting
4    WHERE NRuns > 50
5    UNION
6    SELECT PID AS PlayerID
7    FROM bowling
8    WHERE Nwickets >= 3;
```

Result Grid | Filter Rows:

PlayerID
25001
91001
89001
98002

12 [Set Operator / Subquery Intersect]: Find players who both batted AND bowled in MatchID = 2755 using an INTERSECT query or equivalent subquery structure.

```
SELECT PlayerID, FName, LName
FROM player
WHERE PlayerID IN (
   SELECT PID
   FROM batting
   WHERE MatchID = 2755 )
AND PlayerID IN (
   SELECT PID
   FROM bowling
   WHERE MatchID = 2755 );
```

```
1 •  USE anmol_11;
2 •  SELECT PlayerID, FName, LName
3    FROM player
4    WHERE PlayerID IN (
5        SELECT PID
6        FROM batting
7        WHERE MatchID = 2755 )
8    AND PlayerID IN (
9        SELECT PID
10       FROM bowling
11       WHERE MatchID = 2755 );
```

Result Grid | Filter Rows: | Edit:

PlayerID	FName	LName
94002	Chaminda	Vaas
92002	Muthiah	Murlitharan
91001	Sanath	Jayasuriya
23001	Yuvraj	Singh
98002	Harbhajan	Singh
NULL	NULL	NULL

13 [Complex Grouping & HAVING]: Display Ground, total runs scored on that ground, and total matches played, considering only grounds where more than 2 matches took place and total runs exceed 100.

SELECT M.Ground,

SUM(B.NRuns) AS Total_Runs,

COUNT(DISTINCT M.MatchID) AS Total_Matches

FROM odimatch M

JOIN batting B ON M.MatchID = B.MatchID

GROUP BY M.Ground;

```
1 •  USE anmol_11;
2 •  SELECT M.Ground,
3            SUM(B.NRuns) AS Total_Runs,
4            COUNT(DISTINCT M.MatchID) AS Total_Matches
5    FROM odimatch M
6    JOIN batting B ON M.MatchID = B.MatchID
7    GROUP BY M.Ground;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Ground	Total_Runs	Total_Matches
Brisbane	227	1
Colombo	153	1